Ideal gas, molar mass , flows right in a long straight smooth horizontal adiabatic pipe of area . Molar internal energy at absolute is ; is the gas constant.

A fixed mid-pipe heater supplies constant power , Fig. . Neglect its resistance to flow. In steady flow the state is nonuniform near the heater, becoming uniform far away. In the uniform steady region on the left, pressure is , temperature , and rightward speed . On the right, steady pressure is . Find that region's temperature .

Use a porous plug instead, Fig. . Left steady temperature and pressure are , , with speed rightward and right pressure . With no energy exchange of any kind between gas and plug, find the downstream flow speed .



Fig. Final

V. points

Prism has apex angle below . A ray in the paper plane hits at at any angle. After one refraction it hits , It can reflect totally at . The critical angle at is , and is long enough. Find the range of prism apex for the following two cases:

The ray enters only below the normal;

The ray enters only above the normal.

Fig. Final

VI. points

Charge gives an electric field at distance of magnitude , with constant . A long straight wire carrying produces a field at distance far below its length , of magnitude , with constant .

Fig. Final

Measure in the shown setup using low-frequency hundreds of hertz electric and magnetic fields. are horizontal square plates of capacitance . Side gap far below . is fixed; hangs on one balance hook. are parallel horizontal thin metal rods: length , gap . Rod is fixed; hangs on the other hook. is a known capacitance; is a switch. AC voltage varies with time as ; is frequency. Wire all parts as shown. The balance is level with open, but loses balance when powered. By